

Madiha Khan

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience in SQL, Python, ETL/ELT, data pipelines, data transformation, data modeling, and cloud data platforms. Skilled in Microsoft Fabric, Snowflake, SQL Server, Pandas, NumPy, Power BI, DAX, Power Query, and Advanced Excel. Experienced in building reliable data pipelines, optimizing data workflows, and preparing high-quality datasets to support analytics and data-driven decision-making.

TECHNICAL SKILLS

- Programming Languages & Databases:** Python, SQL, MySQL, SQL Server
- Data Engineering:** ETL/ELT, Data Pipelines, Data Ingestion, Data Transformation, Data Cleaning, Data Validation, Data Modeling, Lakehouse Architecture
- Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, LangChain
- Visualization & BI:** Power BI, DAX, Power Query, KPI Dashboards, Looker Studio
- Tools, Cloud & Data Platforms:** Advanced Excel, Pivot Tables, VLOOKUP, Jupyter Notebook, Streamlit, Git, GitHub, VS Code, MySQL Server, Microsoft Fabric, Snowflake

PROJECTS

Healthcare Data Engineering & Analytics | Microsoft Fabric, SQL, Power BI, DAX, Power Query

- Processed and transformed hospital healthcare data using Microsoft Fabric for analytics and reporting.
- Performed data ingestion, cleaning, preprocessing, and transformation to prepare reliable datasets.
- Used SQL including JOINS, CTEs, GROUP BY, aggregate functions, and window functions for data extraction and analysis.
- Applied data modeling and transformation techniques in Microsoft Fabric to prepare data for downstream analytics.
- Developed and optimized data pipelines in Microsoft Fabric to automate data ingestion and transformation workflows.
- Created and maintained structured data models to support Power BI dashboards, reporting, and business analysis.
- Validated data quality and consistency across transformation stages, ensuring accurate and reliable healthcare datasets for analytics.

Sales Data Engineering & Analytics | Microsoft Fabric, Python, Pandas, Matplotlib, Seaborn

- Ingested and processed 10K+ sales records using Microsoft Fabric and Python.
- Performed data cleaning, transformation, preprocessing, and validation using Pandas.
- Conducted EDA to identify sales and customer trends and generated visual insights using Matplotlib and Seaborn.
- Built and managed data ingestion workflows in Microsoft Fabric to load and transform sales data for analytics.
- Applied feature engineering and aggregation techniques using Pandas to create analysis-ready datasets and business metrics.
- Validated transformed data and analyzed key sales KPIs, including revenue, customer performance, product trends, and regional sales patterns.

AI-Powered Data Assistant using LLMs | Python, LangChain, Groq API, Streamlit

GitHub Repo | Live Demo

- Developed a Generative AI-powered data assistant using LLMs for natural language querying and automated insights generation from CSV datasets.
- Integrated Groq API and LangChain to enable conversational AI, intelligent data exploration, and analytical reporting workflows.
- Built and deployed a Streamlit application with chatbot functionality, file uploads, dynamic visualizations, and prompt-engineering capabilities.

House Price Prediction using Machine Learning | Python, Scikit-learn, Streamlit

- Developed and trained a machine learning regression model on 15K+ real estate records for property price prediction.
- Performed feature engineering, preprocessing, model evaluation, and predictive analytics using Scikit-learn.
- Built and deployed an interactive Streamlit application for real-time property price estimation and decision support.

EDUCATION

B.Tech in Electronics and Telecommunication Engineering

2022 – 2025

R.T.M.N.U University, Nagpur

Diploma in Computer Science Engineering

2019 – 2022

M.S.B.T.E, Nagpur

ADDITIONAL

Achievements: Microsoft Certified: Fabric Data Engineer Associate | July 2026 | Stanford Code in Place Program

(2026) – Stanford University

Languages: English, Hindi